INCH-POUND

MIL-C-24643/7C
14 March 1994
SUPERSEDING
MIL-C-24643/7B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LSMHOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor - AWG size 16, class K stranding, uncoated.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

* Third - Ethylene propylene rubber or cross-linked polyethylene insulation. Minimum average wall thickness 0.020 inch. Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.

Fourth - The specified number of conductors (see table I) cabled together in one or more layers, all in left hand direction, with the minimum amount of fillers necessary, to obtain a firm well-rounded assembly. Cabling sequence to be consecutive, starting with no. 1, from center outward. The length of lay shall be determined to permit a completed cable to meet the twisting and bending endurance tests. A single conductor shall not be used as center for the cable core.

Fifth - Binder tape applied helically with overlap.

* Sixth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.060 inch.

AMSC N/A FSC 6145

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TABLE I. Details

Military part no. M24643/7	Type and size	Number of conductors	Overall Diameter min (inches)	Overall Diameter max (inches)	Bending endurance roller diameter (maximum) (inches)
-01UN	LSMHOF-7	7	0.465	0.500	2.50
-02UN	LSMHOF-10	10	.540	.585	2.50
-03UN	LSMHOF-14	14	.585	.635	3.00
-04UN	LSMHOF-19	19	.650	.705	3.00
-05UN	LSMHOF-24	24	.735	.795	3.50
-06UN	LSMHOF-30	30	.775	.835	4.00
-07UN	LSMHOF-37	37	.855	.925	4.00
-08UN	LSMHOF-44	44	.925	1.000	4.50
-09UN	LSMHOF-61	61	1.100	1.175	5.00

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	4.59
Voltage withstand - volts, root mean square, minimum..	2000
Insulation resistance - megohms/1000 feet, minimum....	500
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
* Group B:	
Cross-linked proof test (percent, maximum)	
Insulation.....	50
Jacket (When tested at 200°C).....	50
Tear - pounds per inch thickness, minimum (ASTM D 470).....	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum.....	700
Elongation - percent, minimum.	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum..	1300
Elongation - percent, minimum.....	160

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EXAMINATION AND TESTS (continued):

	Requirements
Twisting and bending endurance -	
At 75°C, cycles, minimum	6000
At minus 30 ±2°C, cycles, minimum	2000

Group C:

Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
* Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
* Permanence of printing (jacket) - cycles, minimum	125
* Permanence of printing (conductor) - cycles, minimum	25
Heat distortion - percent of unaged, maximum	30

Group D:

* Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18

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QUALIFICATION INSPECTION (continued):

	Requirements
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH:

Size	Feet (nominal)
7 and 10	1500
14 through 61	1500

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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